

MT3144 - Ordinary Differential Equations

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Chapter 1

Introduction

Ordinary differential equations (ODEs) are a mathematical model for Dynamics. Examples:

- Positive Feedback: $\frac{dx}{dt} = kx, k > 0$
- Negative Feedback: $\frac{dx}{dt} = -kx, k > 0$
- Hooke's Law: $\frac{d^2x}{dt^2} = -kx, k > 0$
- Law of Gravity: $\frac{d^2x}{dt^2} = -\frac{kx}{|x|^3}$
- Lotka-Volterra Equations: $\frac{dx}{dt} = kx - xy, \frac{dy}{dt} = -ky + xy$

1.1 Most Important ODE - $x'(t) = f(t)$

Domain of f, u is $\mathbb{R}$ (or an interval of $\mathbb{R}$). We will consider the initial value problem (IVP) for this ODE:

$$\begin{aligned}x'(t) &= f(t) \\x(t_0) &= x_0\end{aligned}$$

Theorem 1.1. $u : I \rightarrow \mathbb{R}$, u is differentiable on I and $u'(t) = f(t)$ for all $t \in I$. Then there exists a unique solution u to the IVP on I :

$$u(t) = x_0 + \int_{t_0}^t f(s)ds \tag{1.1}$$

Proof. Existence: The existence of the solution is guaranteed by the Fundamental Theorem of Calculus.

Uniqueness: Suppose $v : I \rightarrow \mathbb{R}$ is another solution to the IVP. Let $w(t) = u(t) - v(t)$. Then $w'(t) = u'(t) - v'(t) = f(t) - f(t) = 0$. Therefore, $w(t) = C$ for some constant C . Since $w(t_0) = u(t_0) - v(t_0) = x_0 - x_0 = 0$, we have $C = 0$. Thus, $w(t) = 0$ for all $t \in I$, which implies $u(t) = v(t)$ for all $t \in I$. Hence, the solution is unique. $\square$

1.2 Quadratic First Order ODEs

Consider the IVP:

$$\begin{aligned}(x'(t))^2 &= t \\x(0) &= 1\end{aligned}$$

The solution to this IVP is done by taking multiple cases:

- **Case 1:** $x'(t) = \sqrt{t}$

$$\begin{aligned}x(t) &= \int \sqrt{t} dt \\&= \frac{2}{3}t^{3/2} + C\end{aligned}$$

Using the initial condition $x(0) = 1$, we find $C = 1$. Therefore, the solution for Case 1 is:

$$x(t) = \frac{2}{3}t^{3/2} + 1 \quad (1.2)$$

- **Case 2:** $x'(t) = -\sqrt{t}$

$$\begin{aligned}x(t) &= \int -\sqrt{t} dt \\&= -\frac{2}{3}t^{3/2} + C\end{aligned}$$

Using the initial condition $x(0) = 1$, we find $C = 1$. Therefore, the solution for Case 2 is:

$$x(t) = -\frac{2}{3}t^{3/2} + 1 \quad (1.3)$$